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**Screening of Natural Compounds as Potential JAK Inhibitors against Atopic Dermatitis: A
Molecular Docking and Molecular Dynamics Simulation Study**

A Project report submitted to St. Xavier's College (Empowered Autonomous
Institution)

For the partial fulfillment of the Post Graduate Diploma course in
Bioinformatics

By

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Post Graduate Diploma in

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DEPARTMENT OF BIOTECHNOLOGY

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CERTIFICATE

This is to certify that Harshita Prakash Bhagat, student of Post Graduate Diploma in Bioinformatics, at Post Graduate Department of Biotechnology, St. Xavier's College (**Empowered Autonomous Institution**) has submitted the project work titled **Screening of Natural Compounds as Potential JAK Inhibitors: A Molecular Docking and Molecular Dynamics Simulation Study** for the partial fulfillment of the Post Graduate Diploma course in Bioinformatics, during the academic year 2025-2026.

Dr. Norine Dsouza,

Asst. Professor and Head, Department of Biotechnology,

Coordinator, Post Graduate Diploma in Bioinformatics

DECLARATION

I, Harshita Prakash Bhagat, hereby declare that the work described in this report titled, “**Screening of Natural Compounds as Potential JAK Inhibitors: A Molecular Docking and Molecular Dynamics Simulation Study**” submitted to the Department of Biotechnology, St. Xavier’s College (Empowered Autonomous Institution), Mumbai for the partial fulfillment of Post Graduate Diploma in Bioinformatics.

This work is original, conceived and has not been submitted in part or full by me for an award of any other degree or diploma from any other university.

Wherever, other sources of information and contributions of others are involved, every effort is made to indicate this clearly, with due reference to the literature, and acknowledgement of collaborative research and discussions.

Student’s name and signature: Harshita Prakash Bhagat

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Harshita Prakash Bhagat

Abstract

There is a growing body of literature that recognizes the importance of Janus Kinase (JAK) and its isoforms JAK1, JAK2 and JAK3 inhibitors in curing Atopic Dermatitis (AD). This disease is majorly regulated via the JAK/STAT pathway which plays a crucial role in the manifestation of Atopic Dermatitis by presenting rashes and pruritic like lesions on the dermis. The exact mechanism of this pathway can be understood by focusing on the factors that dictate the occurrence of AD. Weakened Immune system, Genetics, Epigenetics and Skin Microbiota are the key drivers of this mechanism. They release of Interferons and Interleukins especially; IL-13, IL-31 and IL-4 along with TSLP by the Th1 and Th2 pathway. In this study, natural compounds were taken into consideration as potential JAK inhibitors in place of FDA-approved JAK inhibitors. The already approved FDA drugs have certain limitations such as a cost of therapy, safety, efficacy and the occurrence of adverse effects. 14 natural compounds were selected on the basis of their ADMET profiles and screened against all three isoforms of JAK. Diosmetin and Naringenin were the top 2 compounds selected against JAK1 (JAK1-DMS, JAK1-NRG) followed by Hecogenin and Diosmetin against JAK2 (JAK2-HEC, JAK2-DMS) and Hecogenin and Verbacoside against JAK3 (JAK3-HEC, JAK3-VBC). These 6 ligand-receptor compounds were subjected to Molecular Docking using UCSF Chimera and AutoDock Vina followed Molecular Dynamic Simulation analysis for 20 ns using GROMACS. On the basis of Protein and Ligand RMSD, RMSF, Radius of Gyration, Surface Accessible Area and number of Intermolecular bonds along with Secondary Special Elements (SSE), Hecogenin displayed structural stability and compact conformational binding with JAK2 and JAK3 along with Verbacoside with JAK3. Hecogenin was considered as a suitable, stable and conformationally accurate target for pan-JAK inhibition activity from the MD Simulation data. Collectively, these findings identify Hecogenin and Verbacoside as the most promising naturally derived lead compounds targeting the JAK-STAT pathway in atopic dermatitis, providing a strong computational foundation that warrants further experimental validation through in vitro enzyme inhibition assays, cell-based studies, and in vivo preclinical evaluation to confirm their therapeutic efficacy and safety as potential natural JAK inhibitors.

Keywords: Atopic Dermatitis, JAK/STAT, Hecogenin, Verbacoside

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1. Introduction

1.1 ATOPIC DERMATITIS – ECZEMA

Atopic Dermatitis (AD) also known as eczema is an inflammatory skin disease manifested by dry skin, erythematous rash, lesions and intense pruritus. Atopic dermatitis is one of the most common chronic inflammatory skin disorders, affecting approximately 220 million people worldwide according to the data collected by the WHO (World Health Organization) Global Burden of Disease Initiative. Originally viewed as a skin disorder of childhood, AD has an estimated prevalence of 20% in children and 10% in adults. (Deckers et al., 2012) The burden of disease ranks number one among nonfatal skin diseases globally, significantly impairing the quality of life of affected individuals and their families. For most patients, AD develops during childhood and is triggered by irritants, allergens, and microbes. Some patients outgrow their condition by adolescence, while others experience localized recurrences after longer periods of resolution in adulthood. In rare instances, AD can also develop later in adult life. (Silverberg & Hanifin, 2013) The distribution and manifestation of AD is actively age-dependent. Genetics, Epigenetics, dysregulated immune system, altered skin biome and environmental impact play a crucial role in shaping the occurrence of AD at different stages of life. In addition to these complexities, AD can also hamper daily life chores as it negatively impacts work and education specially for younger children.

The skin lesions associated with atopic dermatitis are characterized by papules, oozing vesicles on red swollen skin (although redness is more difficult to discern in darker skin and may not always be apparent), crusting, and scaling. These lesions can merge into larger areas, sometimes as a result of extensive scratching. When lesions become chronic, they may result in thickening of the skin, known as lichenification. Atopic dermatitis is a highly variable skin disorder with a wide spectrum of clinical manifestations, and typically fluctuates in severity between periods of flares and remissions. In infants, the face, scalp, cheeks, and extensor sides of the arms and legs are frequently involved, but the whole body may be affected in severe cases. With increasing age, skin lesions are commonly found in areas such as the folds of the knees and elbows (flexural), neck, ears, wrists, ankles, and around the eyes. In adolescents and adults, the hands, feet, head, and neck are frequently affected. (Silverberg & Hanifin, 2013)

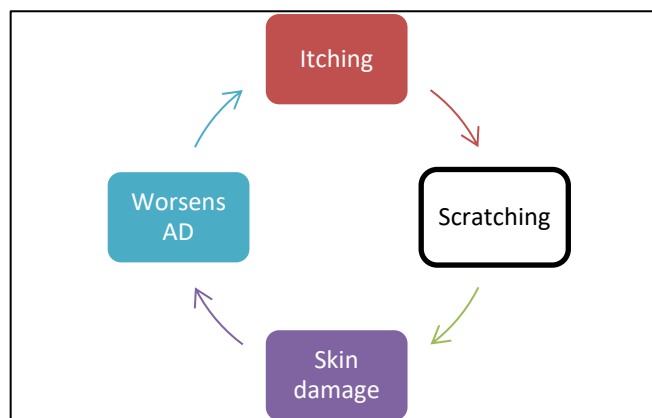


Fig: 1) Eczema Cycle

1.1.1 Pathophysiology of AD

The pathophysiology of AD highlights the skin barrier dysfunction and immune dysregulation as major factors impacting its development. The epidermis plays a crucial role as a physical and functional barrier, and skin barrier defects are the most significant pathologic findings in AD skin. Filaggrin (FLG), transglutaminases, keratins, and intercellular proteins are key proteins responsible for epidermal function. Defects in these proteins facilitate allergen and microbial penetration into the skin. These factors have been used to develop novel therapeutics and preventive strategies against AD. (Kim et al., 2019). Atopic dermatitis patients have a defective skin barrier that is susceptible to xerosis and environmental irritants and allergens that lead to inflammation, pruritus, and the classic clinical findings of atopic dermatitis. The barrier defect may be caused in part by decreased levels of ceramides, which are sphingolipids in the stratum corneum which play a role in the skin's barrier function and prevent trans epidermal water loss. The defective skin barrier allows irritants and allergens to penetrate the skin and cause inflammation via an overactive Th2 response (with increased IL-4, IL-5 cytokines) in acute lesions and Th1 response (with IFN-gamma and IL-12) in chronic lesions. Scratching of the skin also stimulates keratinocytes to release inflammatory cytokines such as TNF-alpha, IL-1, and IL-6. Decreased anti-microbial peptides in the epidermis of atopic patients also contribute to *Staphylococcus aureus* colonization seen in more than 90% of atopic dermatitis patients. *S. aureus* may worsen the inflammation of atopic dermatitis lesions and lead to secondary infection and impetiginized

1.1.1.a. Genetics

The filaggrin (*FLG*) gene is located on chromosome 1q2, and encodes FLG (filaggrin protein), which is a major structural protein in the stratum corneum (SC). Pro-FLG polymers are proteolytically cleaved and dephosphorylated into FLG monomers, which are associated with the aggregation of keratin filaments and the formation of SC. (Kim et al., 2019) The generation of FLG degradation products, urocanic acid and pyrrolidine carboxylic acid, contributes to SC hydration and acidic pH of skin. It is well known that *FLG* null mutations impair skin barrier function and increase the risk of AD. FLG mutations, particularly homozygous mutations, are associated with an increased risk of severe AD with earlier onset, longer persistence, and skin infections. Polymorphisms of various immune pathway genes are associated with an increased risk of AD through alternations in the T-helper (Th) type 2 signaling pathway. Upregulation of interleukin (IL) 4 and IL-13 lowers FLG expression, which leads to skin barrier defects. A gain of functional polymorphisms of type 2 cytokine receptors (IL-4R and IL-13R) are also implicated in AD pathogenesis. Other immune-related genes that contribute to the development of AD include IL-31, IL-33, signal transducer and activator of transcription (STAT) 6, thymic stromal lymphopoietin (TSLP) and its receptors (IL-7R and TSLPR), interferon regulatory factor 2, Toll-like receptor 2, and high-affinity IgE receptor (FcεRI) α gene in specific populations. (Irvine et al., 2011)

1.1.1.b. Epigenetics

Epigenetics are changes in the gene constitution of an individual due to the environmental changes and lifestyle preferences. Epigenetics constitutes cell-specific alterations to the genome that are reversible and heritable and are not directly attributed to the DNA sequence. DNA methylation and the modification of histone proteins that bind to wrapped DNA are two well-known epigenetic mechanisms of genome regulation. The epigenetic marks on the histones include acetylation, phosphorylation, methylation and ubiquitination that affect either the activation or the repression of transcription in a tissue-specific manner. Histone deacetylases (HDACs), histone acetyltransferases (HATs), histone methyltransferases (HMTs) and histone demethylases (HDMs) comprise a large class of enzymes that are involved in catalyzing these modifications in the DNA. (Schmidt & de Guzman Strong, 2021)

methylation more commonly represses transcription by generating a “silencing” mark to the genome. This methylation process is catalyzed by DNA methyltransferases (DNMTs) that transfer a methyl group at the fifth carbon of cytosine resulting in 5-methylcytosine (5mC). 5-mC methylation is typically found at CpG sites, comprised of repeats of cytosine followed by guanine residues, and often near transcriptional start sites (TSS), enhancers and insulators. (Edwards et al., 2017)

1.1.1.c. Skin Microbiome

Human skin hosts a unique microbiota which is well adapted to its hypersaline, acidic environment with limited nutrients. Although it differs greatly from the gut microbiota, it is essential for immune system functioning and prevention of epidermal diseases. The resident bacteria are primarily from the phyla *Actinobacteria*, *Firmicutes*, and *Proteobacteria*, with dominant genera including *Cutibacterium*, *Corynebacterium*, and *Staphylococcus*. These commensals regulate immune responses and enhance skin defense; for instance, *Staphylococcus epidermidis* (*S. epidermidis*) stimulates the production of antimicrobial peptides (AMPs), like beta-defensins, to help prevent pathogenic invasion. (Schmidt & de Guzman Strong, 2021) Human skin hosts 4 distinct microenvironments: moist areas such as the antecubital fossae (dominated by *Staphylococcus*, *Corynebacteria*, and *Proteobacteria*), dry areas such as the volar forearm (with *Flavobacteria*, *Proteobacteria*, and *Corynebacteria*), sebaceous regions such as the upper back (mainly colonized by *Cutibacterium* and *Staphylococcus*), and specialized niches such as dermal layers and hair follicles. The physiological conditions of these microenvironments shape specific microbial communities that thrive in each niche. In addition to bacteria, the skin microbiota includes viral communities like *Polyomavirus* and fungal species such as *Malassezia*, which are prevalent in sebaceous regions and contribute to skin health. (Grice & Segre, 2011) In patients with AD, the balance of skin microbiota is significantly disrupted. This imbalance is characterized by an overgrowth of *S. aureus*, a key contributor to the pathogenesis of AD. *S. aureus* colonizes both lesional and non-lesional skin, suppressing the growth of beneficial commensal bacteria such as *S. epidermidis* and *Corynebacterium* by releasing toxins such as α -toxins and β -toxins, which breakdown the AMPs and weaken the skin defense. This colonization exacerbates skin barrier dysfunction, increases inflammation, and elevates the risk of recurrent skin infections. (Koh et al., 2022) In AD the skin barriers are weakened due to mutations in the filaggrin gene loss of function. This mutation weakens the barrier by reducing natural moisturizing factors (NMFs) such as pyrrolidone carboxylic acid and urocanic acid (UCA), disrupting hydration and allowing the penetration of allergens and pathogens.

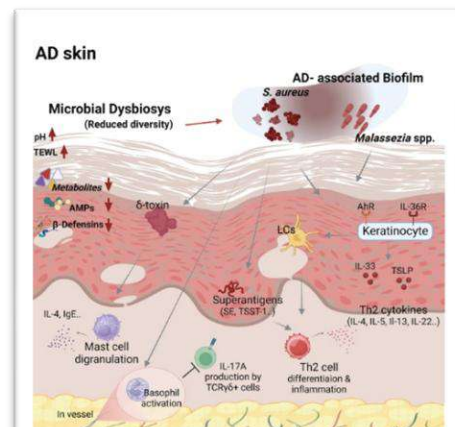


Fig: 2) Immune System Cascade

1.2 JAK/STAT Pathway

The Janus kinase/signal transducer and activator of transcription (JAK/STAT) signaling pathway is regarded as one of the central communication nodes in the cell function. They are responsible for all most all signaling pathways including, hormones, interferons, interleukins and complementary pathways. JAK/STAT-mediated downstream events vary and include hematopoiesis, immune fitness, tissue repair, inflammation, apoptosis, and adipogenesis. Mutations in the JAK/STAT pathway can prove to be fatal for the host. This pathway also has a crucial role in mediating cancer and autoimmune diseases thus, inhibiting it can me promising for treating various diseases. The JAK/STAT signaling pathway is evolutionarily conserved. It is composed of ligand-receptor complexes, JAKs, and STATs. There are 4 members in the JAK family: JAK1, JAK2, JAK3, and TYK2. The STAT family comprises seven members: STAT1, STAT2, STAT3, STAT4, STAT5a, STAT5b, and STAT6. (Hu et al., 2021)

The JAK family: JAK1, JAK2, TYK2, and JAK3

JAK family consists of non-receptor tyrosine protein kinases. When cytokines bind to these receptors, they are activated and in turn activate a signalling cascade which transmits regulatory signals. JAKs have seven homology domains (the JAK homology domain, JH). Starting from the carboxyl terminus, JH1 is the first JH, known as the kinase domain, and is composed of approximately 250 amino acid residues. JH1 encodes a kinase protein that constitutes the kinase structure domain that phosphorylates a substrate; JH2 is a PK domain. JH2 is structurally similar to the kinase domain but has no kinase activity. Its main function is to regulate the activity of the kinase domain. The pseudo kinase domain participates in the interaction of JAK and STAT, and the PK domain can also inhibit Tyr kinase activity by binding to the kinase domain; JH3 with one-half of JH4 constitutes the Src-homology 2(SH2) domain, the combination of one-half of JH4, JH5, JH6, and JH7 constitutes the FERM domain, and the SH2 and FERM domains mainly regulate the binding of JAK and cytokine-receptor membrane-proximal box1/2 regions. (Gurniak & Berg, 1996), (Ferrao & Lupardus, 2017)

1.2.1. JAK1

JAK1 is a conserved tyrosine that constitutes a vital part of the activation loop. The phosphorylation of a double tyrosine in the SH1 domain of each JAK results in a more favorable conformation for substrate binding. JAK1 is present in all tissues and can phosphorylate all STATs JAK1 can promote body hematopoietic function after being activated by IL-3, IL-5, IL-7, granulocyte–macrophage colony-stimulating factor (GM-CSF), or granulocyte colony-stimulating factor (G-CSF). (Ihle et al., 1995)

1.2.2. JAK2

JAK2 also consists of conserved tyrosine sites. Similar to JAK1, JAK2 can also be phosphorylated by members of the gp130 receptor family and class II cytokine-receptor family. It also participates in the signal transduction of the IL-3 receptor family (IL-3R, IL-5R, and GM-CSF receptor), and single-chain receptors (such as erythropoietin receptor (EPO), growth hormone (GH) receptor, prolactin receptor, and thrombopoietin (TPO) receptor). (Schindler & Strehlow, 2000)

1.2.3. JAK3

JAK3 is mainly involved in the signal transduction of the IL-2 receptor, IL-4 receptor, IL-7 receptor, IL-9 receptor, IL-15 receptor, and IL-21 receptor. Lack of JAK3 may lead to autosomal recessive combined immunodeficiency, indicating that JAK3 plays an important regulatory role in the negative selection of T cells and the maintenance of the normal phenotype and function of peripheral T-cells. (Saijo et al., 1997)

1.3 JAK-STAT signaling pathways in AD

In AD, epithelial cell-derived cytokines such as IL-25, IL-33, and thymic stromal lymphopoietin (TSLP) promote the overexpression of Th2 cytokines, including IL-4, IL-5, IL-13, and IL-31, either directly through Th2 cytokine-secreting cells or indirectly through dendritic cell-mediated polarization. IL-4 performs its biological function by binding to an IL-4 receptor (IL-4R), which is a heterodimeric receptor complex on the cellular surface. IL-4R has two subtypes. Type I IL-4R comprises an IL-4 receptor alpha (IL-4R α) chain and a common gamma (γ) chain, whereas type II IL-4R comprises an IL-4R α and an IL-13 receptor alpha 1 (IL-13R α 1) chain. (Brandt & Sivaprasad, 2011) Considering IL-4R α as a functional component responsible for intercellular signal transduction, IL-13 may deliver its signal by binding type II IL-4R with the co-expression of the IL-4R α and IL-13R α 1 chains. (Zurawski et al., 1995) In the IL-4 mediated immune response, IL-4 binds to type I IL-4R, stimulating the phosphorylation of JAK1 and JAK3, which in turn activate and phosphorylate IL-4R α and STAT6. The phosphorylated STAT6 subsequently dimerizes and acts as a transcription factor by binding to specific DNA sequences of IL-4-responsive genes. Activated JAK1 can also phosphorylate Fes kinase, which then phosphorylates insulin receptor substrate-1/2 (IRS-1/2), a protein on the IL-4R α chain linked to several downstream signaling pathways. Phosphorylated IRS-1/2 recruits p85 regulatory subunits, which activate the phosphatidylinositol 3-kinase (PI3K)–protein kinase B (AKT) pathway. The PI3K–AKT pathway then promotes the proliferation, regeneration, and apoptosis of keratinocytes. Conversely, IL-4 and IL-13 can bind to type II IL-4R, inducing the robust phosphorylation of JAK1 and TYK2, followed by the activation and phosphorylation of STAT3 and STAT6. This results in the downregulation of FLG expression and skin barrier dysfunction, and the increased production of TSLP, IL-25, and IL-33 in keratinocytes. (Harb & Chatila, 2020). During the chronic phase of AD, Th17 and Th22 cytokines cooperatively modulate local inflammation through upregulation of proinflammatory cytokines that stimulate epidermal hyperplasia. (Koga et al., 2008) In contrast with the classic Th1 and Th2 immune response through the JAK–STAT signaling pathway, Th17-derived IL-17 recruits ACT1, an IL-17 receptor complex adaptor protein, thus engaging with tumor necrosis factor receptor-associated factors (TRAF6) and activating nuclear factor (NF)- κ B, CCAAT/enhancer binding protein (C/EBP), and MAPK pathways. (Gaffen, 2009) IL-22 heterodimeric receptor complex comprises the IL-22 receptor (IL-22R1) and IL-10 receptor 2 (IL-10R2) subunits, which are expressed in small amounts by keratinocytes of the skin, gut, and lungs. (Koga et al., 2008) After binding to IL-22R1, IL-22 induces the phosphorylation of JAK1 and TYK2. Although STAT3 is considered to be the primary mediator in this cascade, the activation of STAT1 and STAT5 has also been observed in IL-22 signaling. (Lejeune et al., 2002) Higher levels of Th1 cytokines including interferon-gamma (IFN- γ), IL-12, and granulocyte colony stimulating factor (G-CSF) are found in chronic AD lesions. In the pathogenesis of AD, IL-12 secreted by dendritic cells can polarize Th1 cells from naïve T-helper cells and inhibit the activation of Th2 cells. The IL-12 signaling pathway is initiated by binding to a distinct heterodimeric receptor comprising interleukin 12 receptor beta (IL-12R β)1 and IL-12R β 2 subunits. Subsequently, JAK2 and TYK2 are polarized, resulting in the mediation of signal transduction through the activation of STAT4, and to a lesser degree STAT1, STAT3, and STAT5. IFN- γ contributes to epidermal dysfunction by reducing the expression levels of ceramides and long-chain fatty acids. (Bastian et al., 2019)

Category	Name	Plant/Source	Pathway	Mechanism
Flavonoids	Liquiritigenin	Glycyrrhiza uralensis	F-κB and MAPK pathways.	Decreased IL-2 production and CD69 expression on the surface of stimulated cells.
	Naringenin	Peels of citrus and grapes,	M1 marker CD68 and modulated HMGB1, RAGE, ERK1/2 and NF-κB p65 expression	Skin repair by switching the macrophage phenotype from M1 to M2.
	Diosmetin	Peel of citrus lemon	JAK/STAT signalling induced by IL-4 but also suppressed the phosphorylation of MAP kinases .	Levels of proinflammatory cytokines in skin lesions, such as IL-4, TNF-α, IL-1β and IL-6.
	Baicalein	Scutellariae radix	Inhibiting the production of IL-4, IL-6 and TNF-α and inducing the expression of IFN-γ, regulation of keratinocyte differentiation and proliferation.	Balance between Th1 and Th2 cells
	Quercetin	Fruits and vegetables	Acts by regulating the TLR2 and JAK2/STAT3 pathways and inhibiting STAT3 tyrosine phosphorylation in inflammatory cells .	Regulation of mast cells, keratinocytes, and Th1/Th2 cells.
Alkaloids	Piperine	Black pepper	Inhibitor of STAT6, can inhibit trimellitic anhydride (TMA)-induced AD-like Th2-related immune responses via the STAT6/GATA3/IL-4 signaling pathway.	PSE strongly blocked NF-κB and MAPKs
Terpenoids	Astaxanthin	Xanthophyll carotenoid	iNOS and COX-2 and NF-κB activity by blocking the expression of the NF-κB subunits p65 and p50.	AST can treat AD by inhibiting the NF-κB signaling pathway, regulating nitric oxide and inhibiting oxidation.
	Fucoxanthin	Microalgae,	Inhibit the expression of IL-33 derived from keratinocytes, recruit eosinophils, and stimulate the expression of TGF-β required for tissue remodelling.	Inhibit TB positivity and degranulation of BMMCs in a dose-dependent manner, thus inhibiting mast cell-dependent scratching behaviour.
	Igalan	Inula heleniu	inhibit the activation of NF-κB and STAT1/STAT3 in TNF-α/IFN-γ-	Resulting in the downregulation of the expression of inflammatory genes such as MDC/CCL22, TARC/CCL17, and RANTES/CCL5, promote keratinocyte structural components and differentiation genes .
	Elemol	Chamaecyparis obtusa leaves	The genes expression of T cell-derived cytokines (TNF-α, IL-1β, IL-6, and IκBα) in the AD-like lesions was significantly blocked by elemol.	Ameliorated the hyper-allergic reaction in the mouse model of AD by attenuating serum IgE levels and decreasing mast cell infiltration into the dermis.
	Taxifolin glycoside	Rhododendron mucronulatum	Downregulated IgE levels, a diagnostic indicator of AD, and Th2 cytokine levels, including IL-4, IL-5, and IL-13.	Immunoregulatory function and anti-inflammatory activity.
	Hecogenin	Agave sisalana	Reduced the production levels of TNF-α and IL-12 in AD-like	

			inflammatory skin.	
	Verbascoside	Plantago Asiatica L.	Abolish the upregulation of the costimulatory molecules CD86 and CD54 and secretion of the proinflammatory cytokines TNF- α and IL-6 and the activation of the NF- κ B pathway.	
Others	Ferulic acid	Angelica sinensis,	Inhibited Th2 cytokines (IL-4, IL-13, and IL-31) and IgE levels and the mRNA expression of proinflammatory cytokines (IL-6, IL-4 and TNF- α)	

Table: 1) Natural Compounds and their role against JAK/STAT Pathway

1.6. Computational Methods in Drug Discovery

Computer-aided drug design is a crucial part of drug discovery in today's modern science. although progress was being made in CADD, the potential for high-throughput screening (HTS) had begun to take precedence as a means for finding novel therapeutics. This brute force approach relies on automation to screen high numbers of molecules in search of those that elicit the desired biologic response. The method has the advantage of requiring minimal compound design or prior knowledge, and technologies required to screen large libraries have become more efficient. CADD has given us the ability to screen compounds and to eliminate those which do not show desired response thus, saving time and effort before proceeding towards the wet lab studies. CADD is capable of increasing the hit rate of novel drug compounds because it uses a much more targeted search than traditional HTS and combinatorial chemistry. It not only aims to explain the molecular basis of therapeutic activity but also to predict possible derivatives that would improve activity. In a drug discovery campaign, CADD is usually used for three major purposes: (1) filter large compound libraries into smaller sets of predicted active compounds that can be tested experimentally; (2) guide the optimization of lead compounds, whether to increase its affinity or optimize drug metabolism and pharmacokinetics (DMPK) properties including absorption, distribution, metabolism, excretion, and the potential for toxicity (ADMET); (3) design novel compounds, either by "growing" starting molecules one functional group at a time or by piecing together fragments into novel chemotypes. To perform a virtual screen, however, a virtual library must be available for screening. Virtual libraries can be acquired in a variety of sizes and designs including general libraries that can be used to screen against any target, focused libraries that are designed for a family of related targets, and targeted libraries that are specifically designed for a single target. Some of the widely used ligand databases are PubChem, Zinc, DrugBank, LIGAND etc. (Sliwoski et al., 2014), (Song et al., 2009)

1.7. Molecular Docking Studies

Molecular docking is one of the most frequently used methods in structure-based drug design because of its ability to predict, with a substantial degree of accuracy, the conformation of small-molecule ligands within the appropriate target binding site. Following the development of the first algorithms in the 1980s, molecular docking became an essential tool in drug discovery. investigations involving crucial molecular events, including ligand binding modes and the corresponding intermolecular interactions that stabilize the ligand-receptor complex, can be

conveniently performed. Furthermore, molecular docking algorithms execute quantitative predictions of binding energetics, providing rankings of docked compounds based on the binding affinity of ligand-receptor complexes. (López-Vallejo et al., 2011) Molecular docking programs use scoring functions to estimate the binding energetics of the predicted ligand-receptor complexes. The energy variation, due to the formation of the ligand-receptor structure, is given by the binding constant (K_d) and the Gibbs free energy (ΔG_L). Prediction of the binding energy is performed by evaluating the most important physical-chemical phenomena involved in ligand-receptor binding, including intermolecular interactions, desolvation and entropic effects. Therefore, the greater the number of physical-chemical parameters evaluated, the greater the accuracy of the scoring function. (Jain, 2006)

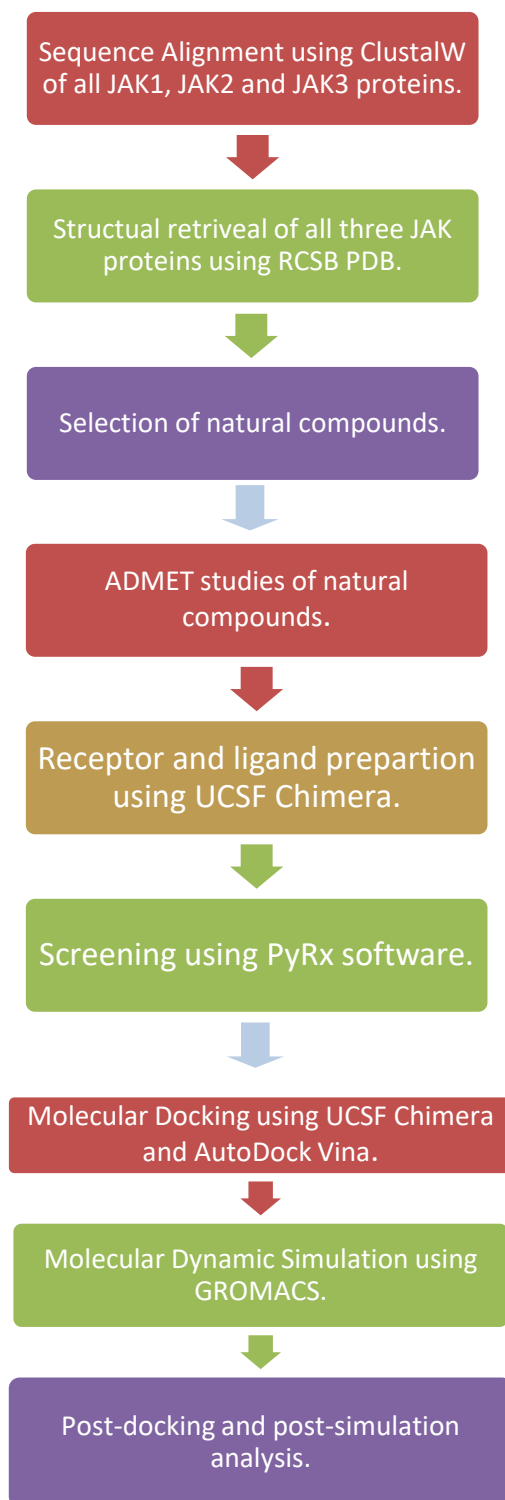
1.8. Molecular Dynamics Simulation

Enzymes and receptors can undergo conformational changes during the molecular recognition process. In some cases, these structural rearrangements are small and the ligand fits in a binding site with little mobility. (Lin, 2011) Such conformational issues can be viewed using MD simulations. Usually, the ligand stabilizes a subset of several possible conformations of the receptor, shifting the equilibrium toward the minimum energy structures. In such cases, MD simulations can produce alternative conformational states corresponding to these ligand-induced structures. Molecular dynamics applies Newton's equations of motion, as described in classical mechanics, to specify the position and speed of each atom in the system under study. As a result, the trajectory and temporal evolution of a ligand-receptor complex can be examined. Initially, a specific configuration is attributed to the atoms with the purpose to reproduce the temperature and pressure of the real system. From the computation of the forces acting on each particle, it is possible to determine the position and velocity of each of these atoms at a posterior time. These calculations are repeatedly performed until the molecular trajectories are integrated for a given time interval. (Nichols et al., 2011)

2. Objectives

1. To perform structural retrieval and sequence comparison of all three JAK family proteins.
2. To perform Natural compounds selections, receptor and ligand preparation
3. To screen using PyRx software
4. To perform Molecular Docking using Chimera and AutoDock Vina
5. To perform Molecular Dynamic Simulation using GROMACS
6. To perform post-docking and post-simulation analysis

3. Methodology Pipeline



4. Results

4.1. Sequence Retrieval and Multiple Sequence Alignment of JAK Family Proteins.

The amino acid sequences of JAK1 (NP_001308782.1), JAK2 (NP_001309125.1) and JAK3 (NP_001427368.1) were retrieved from NCBI protein database. Multiple sequence alignment of the three JAK proteins was performed using MEGA X1 ClustalW sequencing. The Tyrosine-protein kinase 1 (TYK1) domain (position: 583 – 848) was targeted for sequence alignment as it is the binding site for all the ligand molecules. The alignment revealed several conserved residues across all three sequences, including, Valine-Leucine-Leucine-Lysine-Valine (VLLKV), Aspartic Acid (K), Histidine (H), Phenylalanine (F), Glutamic Acid (E), Alanine-Alanine-Serine (AAS), Methionine (M), Serine (S), Histidine (H), Glycine-Valine-Cysteine (GVC). These conserved residues are likely to represent functionally or structurally critical domains within the binding site of JAK family. JAK1 and JAK2 shared 11.33% sequence identity and JAK2 and JAK3 shared 13.07% identity.



Fig: 4) Protein Sequence Alignment

4.2. Protein Structure Retrieval

The three-dimensional crystal structures of JAK1, JAK2, and JAK3 were retrieved from the RCSB Protein Data Bank (PDB) for structural analysis and molecular docking studies. All three structures were experimentally determined using X-ray diffraction crystallography.

The JAK1 structure (PDB ID: 4E5W) was resolved at a resolution of 1.86 Å. Examination of the structure revealed missing residues in chain A (Pro912, Glu913, Ser914, Gly915, Gly916, Asp947, Gly948, Gly949) and in chain B (Gly853–Glu864, Glu913, Ser914, Gly915, Gly916, Asp947, Gly948, Gly949, Thr1030, Asp1031, Lys1154), indicating loop regions that were unresolved in the electron density map.

The JAK2 structure (PDB ID: 3KRR) was resolved at a resolution of 1.80 Å, with minimal missing residues limited to Gly838 and Gly1132 in chain A, suggesting a largely complete and well-resolved structure.

The JAK3 structure (PDB ID: 5LWM) was resolved at the highest resolution of 1.55 Å among the three proteins, with missing residues confined to Ser810, Met811, Gln812, and Asp813 in chain A, indicating a high-quality structure suitable for docking analysis.

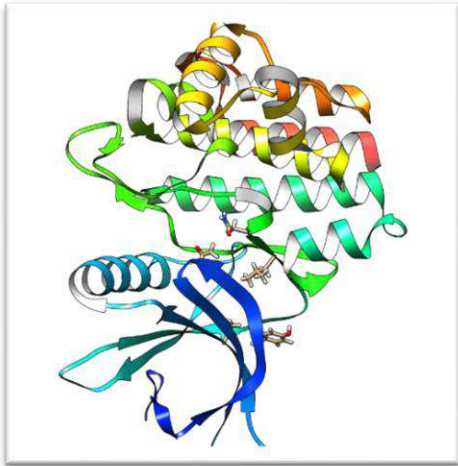
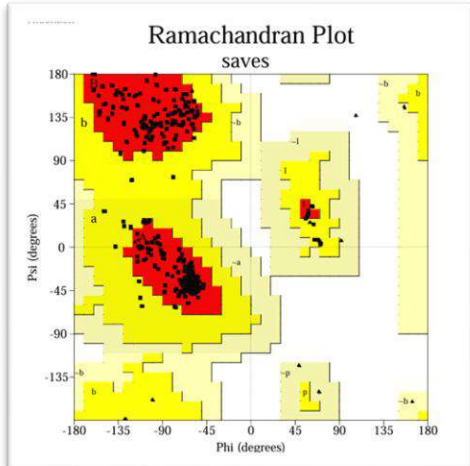
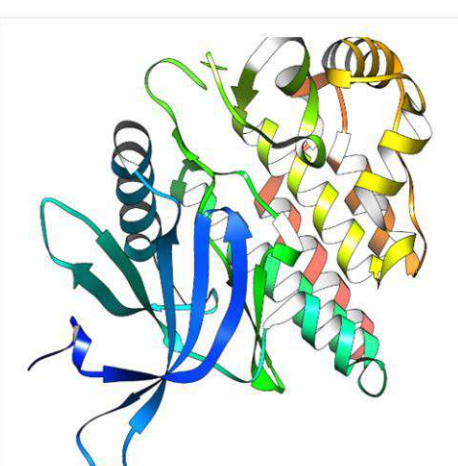
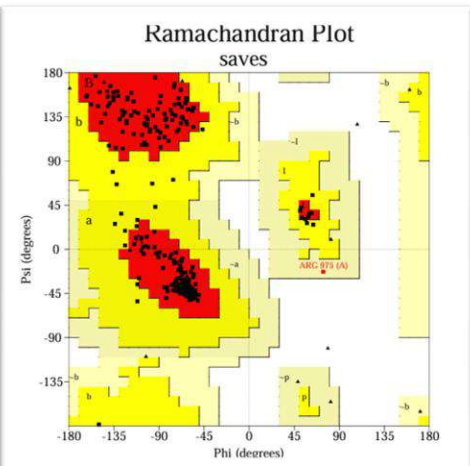
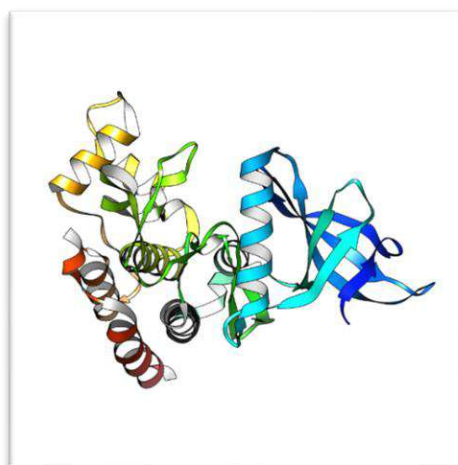
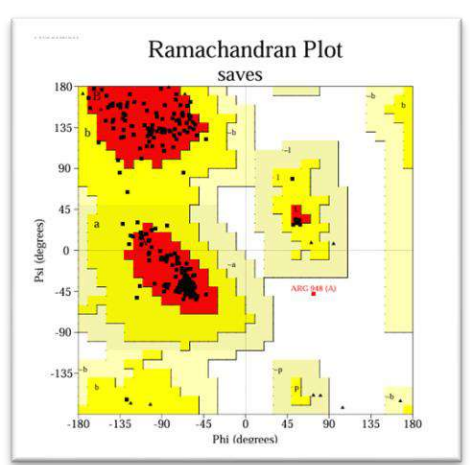
PROTEIN	3D PDB STRUCTURE	RAMACHANDRAN PLOT
JAK1: 4E5W		
JAK2: 3KRR		
JAK3: 5LWM		

Fig 5): Protein Structure Analysis with Ramachandran Plots

ProsaWeb Results

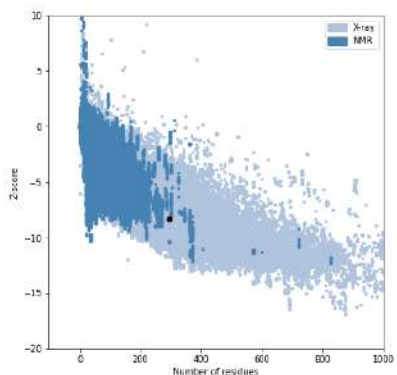
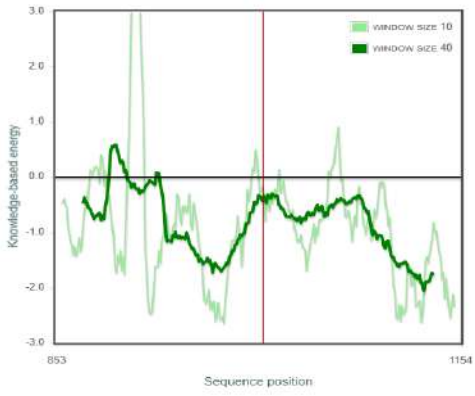
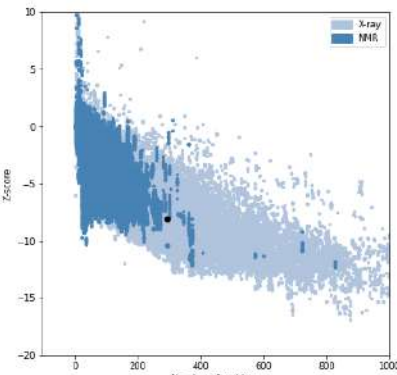
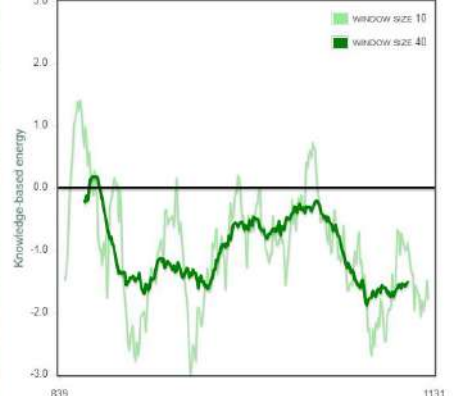
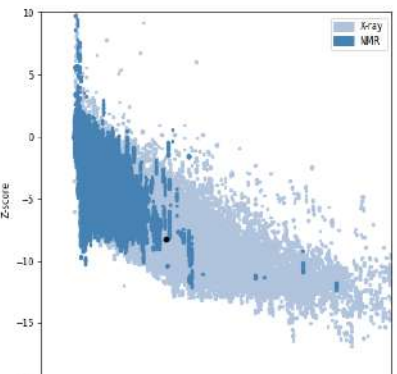
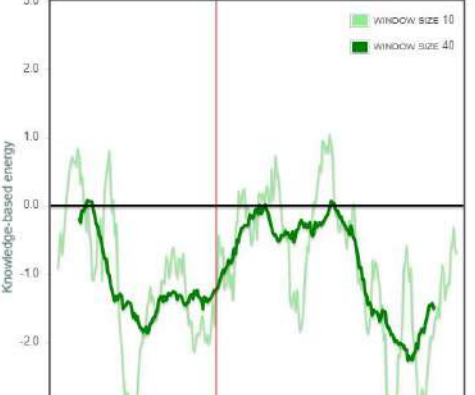
Receptor	Z-score	Z-score Diagram	Model Quality
JAK1	-8.35		
JAK2	-8.09		
JAK3	-8.21		

Fig 6): Protein Structure Analysis: Prosa Web Results

4.3 Selection of the natural compounds:

A total of 14 natural compounds were initially retrieved from the PubChem database. Following application of Lipinski's Rule of Five and ADMET profiling and molecular screening approaches, 6 compounds were shortlisted for molecular docking and MD simulation.

Compound	CID	SMILES
Quercetin	5280343	<chem>C1=CC(=C(C=C1C2=C(C(=O)C3=C(C=C(C=C3O2)O)O)O)O)O</chem>
Liquiritigenin	114829	<chem>C1[C@H](OC2=C(C1=O)C=CC(=C2)O)C3=CC=C(C=C3)O</chem>
Naringenin	439246	<chem>C1[C@H](OC2=CC(=CC(=C2C1=O)O)O)C3=CC=C(C=C3)O</chem>
Diosmetin	5281612	<chem>COC1=C(C=C(C=C1)C2=CC(=O)C3=C(C=C(C=C3O2)O)O)O</chem>
Baicalein	5281605	<chem>C1=CC=C(C=C1)C2=CC(=O)C3=C(O2)C=C(C(=C3O)O)O</chem>
Piperine	638024	<chem>C1CCN(CC1)C(=O)/C=C/C=C/C2=CC3=C(C=C2)OC(=O)C3</chem>
Pseudoephedrine	7028	<chem>C[C@@H]([C@H](C1=CC=CC=C1)O)NC</chem>
Astaxanthin	5281224	<chem>CC1=C(C(C[C@@H](C1=O)O)(C)C)/C=C/C(=C/C=C/C(=C/C=C/C=C(/C=C/C=C(/C=C/C2=C(C(=O)[C@H](CC2(C)C)O)C)\C)/C)/C</chem>
Igalan	14038398	<chem>CC(=C)[C@H]1C[C@H]2[C@@H](C[C@]1(C)C=C)OC(=O)C2=C</chem>
Elemol	92138	<chem>CC(=C)[C@@H]1C[C@@H](CC[C@@]1(C)C=C)C(C)(C)O</chem>
Taxifolin 3-O-beta-D-glucopyranoside	14187089	<chem>C1=CC(=C(C=C1[C@@H]2[C@H](C(=O)C3=C(C=C(C=C3O2)O)O)[C@H]4[C@@H]([C@H]([C@@H]([C@H](O4)CO)O)O)O)O)O</chem>
Hecogenin	91453	<chem>C[C@@H]1CC[C@@]2([C@H]([C@H]3[C@@H](O2)C[C@@H]4[C@@]3(C(=O)C[C@H]5[C@H]4CC[C@@H]6[C@@]5(CC[C@@H](C6)O)C)C)OC1</chem>
Verbascoside	5281800	<chem>C[C@H]1[C@@H]([C@H]([C@H]([C@@H](O1)O)[C@@H]2[C@H]([C@@H](O[C@@H]([C@H]2OC(=O)/C=C/C3=CC(=C(C=C3)O)O)CO)OCCC4=CC(=C(C=C4)O)O)O)O)O</chem>

Ferulic Acid	445858	<chem>COC1=C(C=CC(=C1)/C=C/C(=O)O)O</chem>
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Table 2): SMILES data of 14 compounds

4.4. ADMET studies

Compound	Molecular weight (g/mol)	H-bond acceptor	H-bond donor	Rotatable bonds	Log P	TPSA (Å ²)	Log S	BBA score
Quercetin	302.24	7	5	1	1.63	131.36	-3.16	No
Liquiritigenin	256.25	4	2	1	1.73	66.76	-3.28	Yes
Naringenin	272.25	5	3	1	1.75	86.99	-3.49	No
Diosmetin	300.26	6	3	2	2.47	100.13	-4.06	No
Baicalein	270.24	5	3	1	2.43	90.90	-4.03	No
Piperine	285.34	3	0	4	3.38	38.77	-3.74	Yes
Pseudoephedrine	165.23	2	2	3	1.93	32.26	-1.62	Yes
Astaxanthin	596.84	4	2	10	-	74.60	-	-
Igalan	232.32	2	0	2	2.71	26.30	-3.71	Yes
Elemol	222.37	1	1	3	3.20	20.23	-3.80	Yes
Taxifolin 3-O-beta-D-glucopyranoside	466.39	12	8	4	0.51	20.60	-2.33	No
Hecogenin	430.62	4	1	0	4.11	55.76	-5.55	Yes
Verbascoside	624.59	15	9	11	2.15	245.29	-2.87	No
Ferulic Acid	194.18	4	2	3	1.62	66.76	-2.11	Yes

Table 3): ADMET Profiling

4.5. Screening of Natural Compounds

JAK1 docking analysis

Ligand	Ligand name	Binding Aff	RMSD/UL	RMSD/LB
4e5w_prepared_5281612_uff_E=324.34	Diosmetin	-9.4	0	0
4e5w_prepared_439246_uff_E=195.81	Naringenin	-9.4	0	0
4e5w_prepared_114829_uff_E=185.86	Liquiritrengin	-9.3	0	0
4e5w_prepared_5281800_uff_E=559.11	Verbacoside	-9.1	0	0
4e5w_prepared_5281605_uff_E=241.95	Baicalein	-9	0	0

JAK2 docking analysis

Ligand	Ligand Name	Binding Aff	RMSD/UL	RMSD/LB
3KRR_prepared_91453_uff_E=836.94	Hecogenin	-10	0	0
3KRR_prepared_5281612_uff_E=324.34	Diosmetin	-9.4	0	0
3KRR_prepared_5280343_uff_E=380.43	Quercetin	-9.1	0	0

JAK3 docking analysis

Ligand	Ligand Name	Binding Aff	RMSD/UL	RMSD/LB
5lwm_prepared_5281224_uff_E=748.45	Astaxanthin	-10.1	0	0
5lwm_prepared_91453_uff_E=836.94	Hecogenin	-10	0	0
5lwm_prepared_5281800_uff_E=559.11	Verbacoside	-9.3	0	0
5lwm_prepared_439246_uff_E=195.81	Naringenin	-8.8	0	0

Table 4): PyRx Screening Analysis

4.6.Molecular Docking Analysis

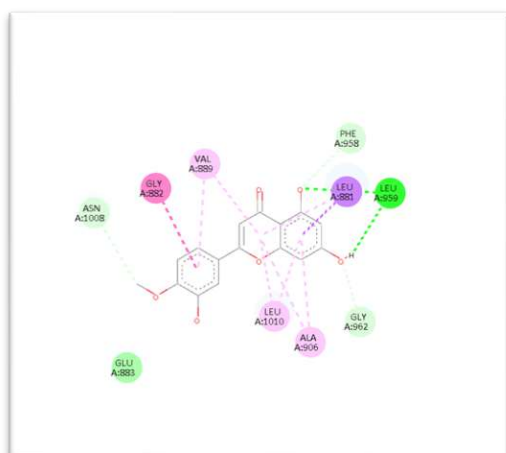
Among the 14 compounds screened, Diosmetin showed maximum affinity of -9.4 kcal/mol against JAK1. Pi-Alkyl bonds were made with *Leu1010*, *Val998* and *Ala906*. Pi-Sigma bond was made with *Leu881* and Amide-Pi Stacked was made with *Gly882* in the ATP binding pocket of the protein. Along with this, Naringenin also showed affinity of -9.3 kcal/mol against JAK1. Pi-Sigma bond was made with *Leu881*, Pi-Alkyl bond was made with *Ala906*, *Val889* and *Leu1010*. An unfavorable acceptor bond was made with *Leu59*. Diosmetin also showed affinity of -9.2 kcal/mol against JAK2. Pi-Alkyl bonds were made with *Leu983* and *Ala880*, Pi-Sigma bonds were made with *Leu855* and *Val863* and a unique Pi-Sulfur bond was made with *Met929*. Hecogenin showed an affinity of -10.01 kcal/mol against JAK2. Pi-

alkyl bonds were made with *Leu983*, *Val863* and *Leu855*. Hecogenin also showed affinity of -10.14 kcal/mol against JAK3. Pi-alkyl bonds were made with *Val836*, *Leu956*, *Cys909* and *Leu828*. Verbacoside displayed an affinity of -9.1 kcal/mol against JAK3. Pi-alkyl bonds were made with *Ala966* and *Val836*.

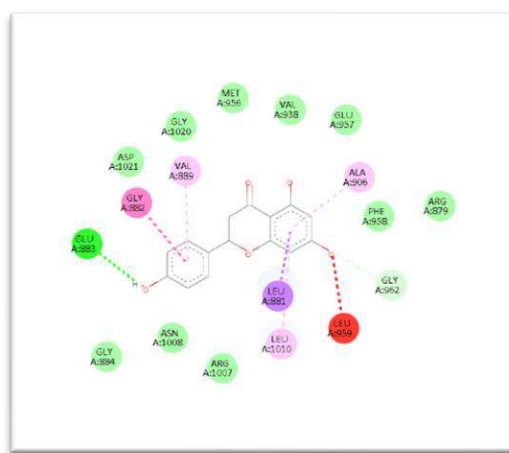
Protein	Ligand	Binding Affinity (kcal/mol)
JAK1	Diosmetin	-9.4
	Naringenin	-9.3
JAK2	Diosmetin	-9.2
	Hecogenin	-10.01
JAK3	Hecogenin	-10.14
	Verbacoside	-9.1

Table 5): Molecular Docking Scores

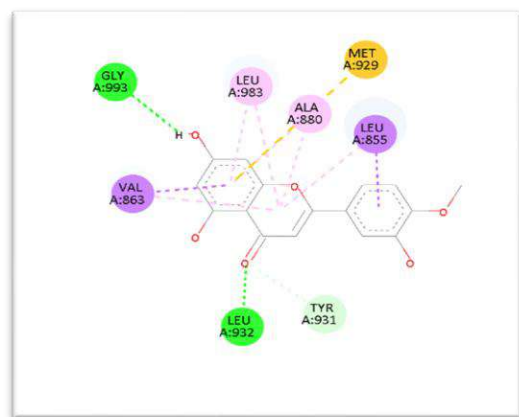
4.7. 2D-Interaction Profiling using BIOVIA Discovery Studio



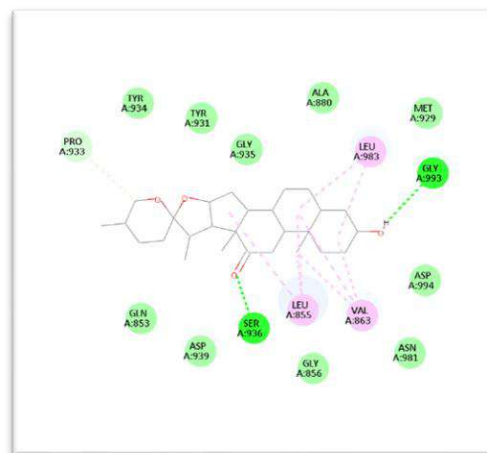
JAK1 with Diosmetin



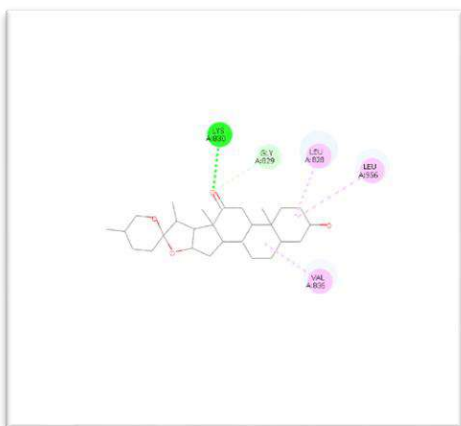
JAK1 with Naringenin



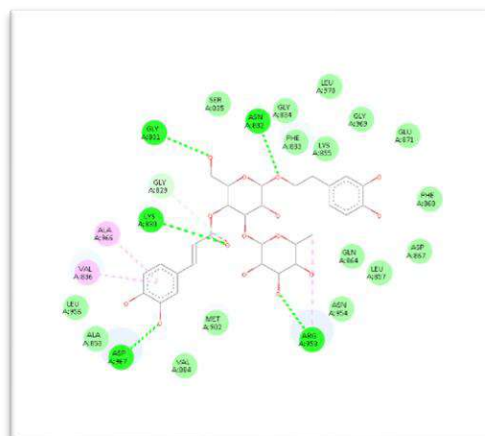
JAK2 with Diosmetin



JAK2 with Hecogenin

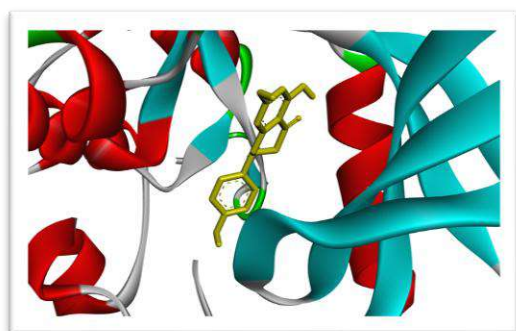


JAK3 with Hecogenin

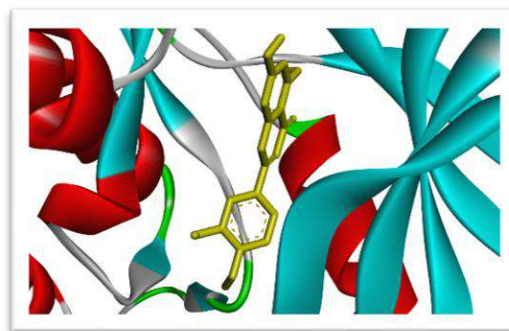


JAK3 with Verbacoside

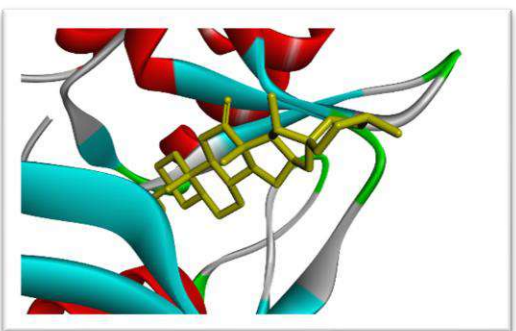
3D Binding pose of Protein and Ligand



JAK1 Naringenin



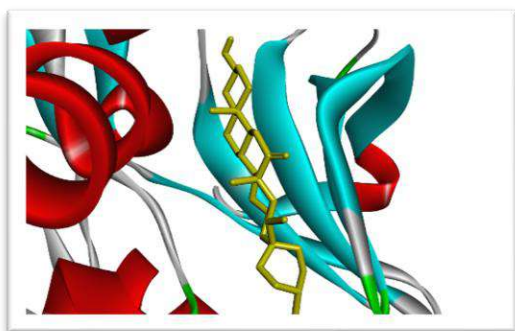
JAK1 Diosmetin



JAK2 Hecogenin



JAK2 Diosmetin



JAK3 Hecogenin



JAK3 Verbaoside

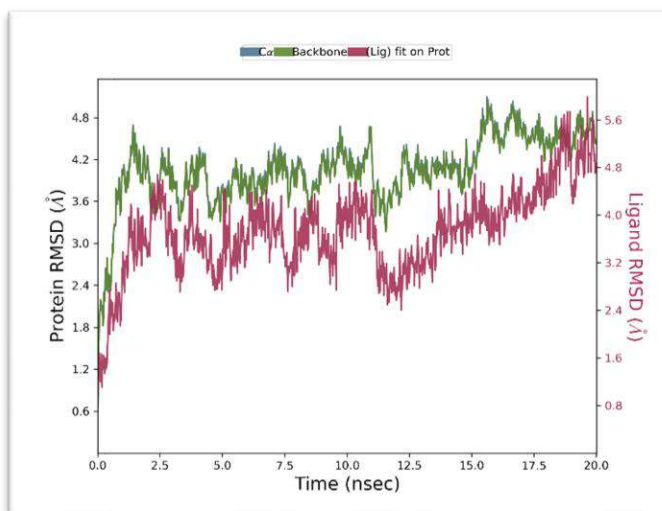
4.8 Molecular Dynamic Simulation

4.8.1 JAK1 Section:

1) JAK1 with Diosmetin

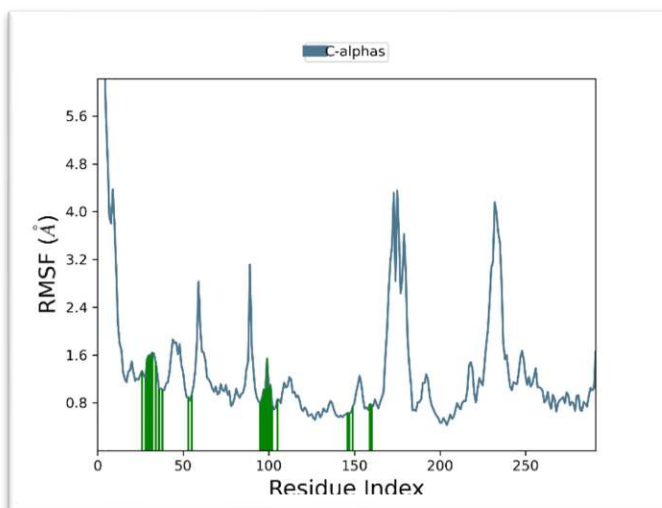
The molecular dynamics simulation of the JAK1-DMS complex was conducted for 20 ns under NPT ensemble conditions at 300K temperature using Desmond. The system comprised of 49544 atoms along with 14920 water molecules and a net charge of zero.

Structural Stability (RMSD): The protein backbone RMSD fluctuated between approximately 1.8 and 4.8 Å throughout the 20 ns simulation. While these values are slightly elevated compared to typical small globular proteins, the RMSD did not exhibit a continuous upward trend and showed fluctuation around a thermal average, indicating that the system reached equilibration. The ligand RMSD (Lig fit Prot) ranged between approximately 0.8 and 5.6 Å without sustained divergence from the protein, indicating that the ligand ($C_{16}H_{28}O_6$, molecular weight 316.398 Da) remained associated with the JAK1 binding pocket throughout the trajectory despite exhibiting some positional flexibility within the active site.

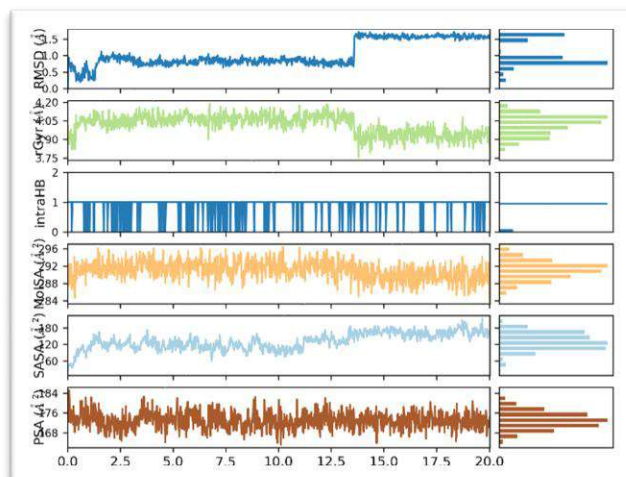


Residue Flexibility (RMSF): Protein RMSF analysis revealed that the majority of residues fluctuated below 1.6 Å, with localized peaks reaching up to 4.8 Å and 5.6 Å observed at loop regions and terminal segments, consistent with the inherent flexibility of unstructured protein regions. Ligand-contacting

residues, marked by green bars at approximately residue indices 30–50, 100, and 145–155, displayed comparatively lower fluctuation values, confirming that the binding site maintained structural stability throughout the simulation.



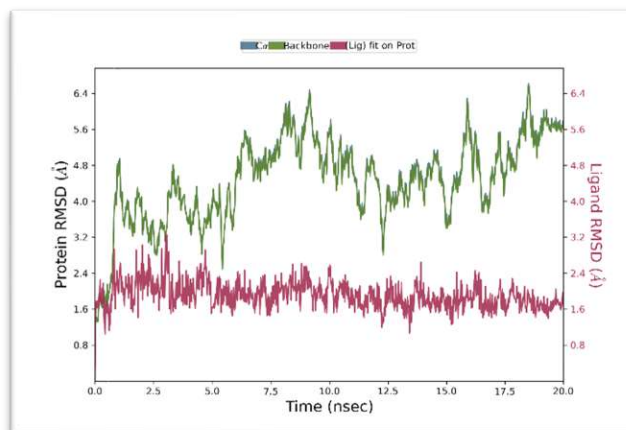
Ligand Properties: Throughout the simulation, the ligand RMSD remained stable between approximately 0.5 and 1.5 Å for the first 12 ns before showing a modest increase to approximately 1.5 Å for the remainder of the trajectory, consistent with a conformational adjustment within the binding site rather than dissociation. The radius of gyration was stable between 3.75 and 4.20 Å, confirming that the ligand maintained a compact conformation throughout the simulation. Intramolecular hydrogen bonds fluctuated between 0 and 1 per frame, indicating that the ligand's hydroxyl groups were predominantly engaged in intermolecular interactions with the protein rather than forming internal bonds. The molecular surface area remained stable between 284 and 296 Å², the solvent accessible surface area ranged from 60 to 180 Å², and the polar surface area was consistently maintained between 168 and 184 Å², all of which reflect a well-embedded ligand conformation within the hydrophilic environment of the JAK1 binding pocket



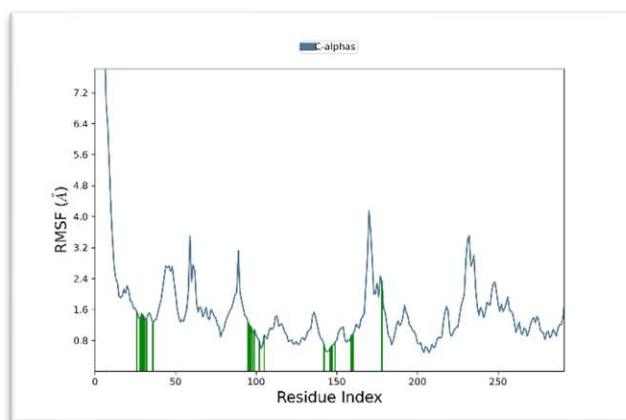
2) JAK1 with NRG

The molecular dynamics simulation of the JAK1-NRG complex was conducted for 20 ns under NPT ensemble conditions at 300K temperature using Desmond. The system comprised of 49537 atoms along with 14919 water molecules and a net charge of zero.

Structural Stability (RMSD): The protein backbone RMSD fluctuated between 0.8 to 6.4 Å throughout the 20 ns simulation. While the RMSD plot displayed an elevation peak reaching 6.4 Å between 0 and 10 ns, the trajectory appeared to be stable later without a risen peak suggesting system reached equilibrium. The ligand RMSD ranged between 0.8 to 1.6 Å.

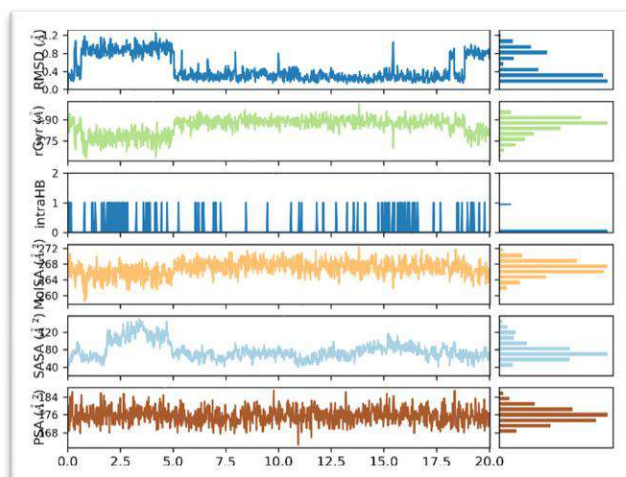


Residue Flexibility (RMSF): Protein RMSF analysis revealed that the majority of residues maintained fluctuations between 0.8 and 2.4 Å, with localized peaks reaching up to 4.8 Å and 7.2 Å at loop regions and terminal segments, consistent with the expected flexibility of unstructured protein regions. Ligand-contacting residues, indicated by green bars at approximately residue indices 30–55, 100, and 145–155, exhibited comparatively low fluctuation values, confirming that the binding site residues maintained structural stability throughout the simulation despite the overall conformational dynamics of the protein.



Ligand Properties: Throughout the simulation, the ligand RMSD remained low and stable between approximately 0.2 and 1.0 Å for the first 12 ns before showing a brief elevation to approximately 0.8–1.2 Å toward the latter part of the trajectory, indicating a minor conformational adjustment within the binding pocket without dissociation from the active site. The radius of gyration was consistently maintained between 3.75 and 3.90 Å throughout the full 20 ns, confirming that the ligand retained a compact and stable three-dimensional conformation during the simulation. Intramolecular hydrogen bonds fluctuated between 0 and 1 per frame, indicating that the ligand's hydroxyl groups were

predominantly directed outward toward protein residues rather than forming internal bonds, which is favorable for intermolecular binding. The molecular surface area remained stable between 260 and 272 Å², the solvent accessible surface area ranged from 40 to 120 Å² with a gradual decrease after 5 ns suggesting progressive burial within the binding pocket, and the polar surface area was consistently maintained between 168 and 184 Å², reflecting a well-embedded polar ligand conformation within the JAK1 active site throughout the simulation.

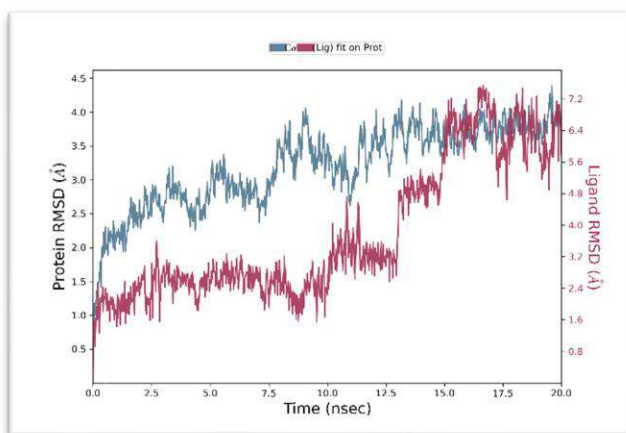


4.8.2. JAK2 Section:

3) JAK2 with DMS

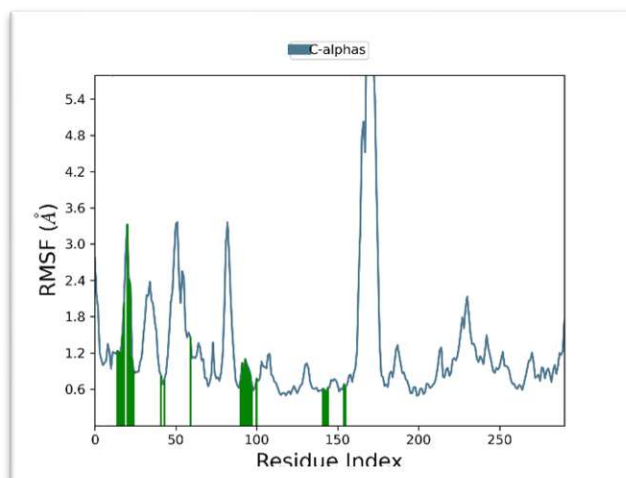
The molecular dynamics simulation of the JAK2-DMS complex was conducted for 20 ns under NPT ensemble conditions at 300K temperature using Desmond. The system comprised of 37935 atoms along with 11033 water molecules and a net charge of zero.

Structural Stability (RMSD): The protein backbone RMSD fluctuated between 1.00 and 4.00 Å throughout the 20 ns simulation. The system reached equilibrium. The ligand remained consistent throughout the simulation with a small peak at 3.5 Å suggesting the system reached equilibrium. The ligand RMSF ranged from 0.5 to 7 Å with peaks at 12.5 ns and 15 ns, with values comparable to or slightly exceeding the protein RMSD at various time points.

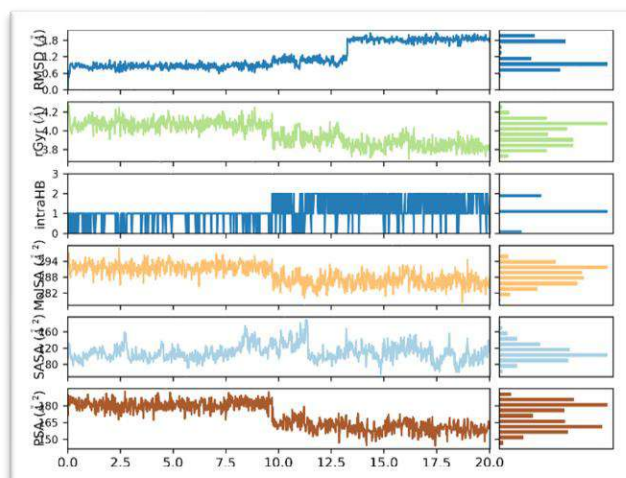


Residue Flexibility (RMSD): Protein RMSF analysis revealed that the majority of residues maintained fluctuations between 1.2 and 2.4 Å, with localized peaks reaching up to 3.0 Å and 5.4 Å at loop regions.

and terminal segments, consistent with the expected flexibility of unstructured protein regions. Ligand-contacting residues, indicated by green bars at approximately residue indices 30–55, 100, and 145–155, exhibited comparatively low fluctuation values, confirming that the binding site residues maintained structural stability throughout the simulation despite the overall conformational dynamics of the protein.



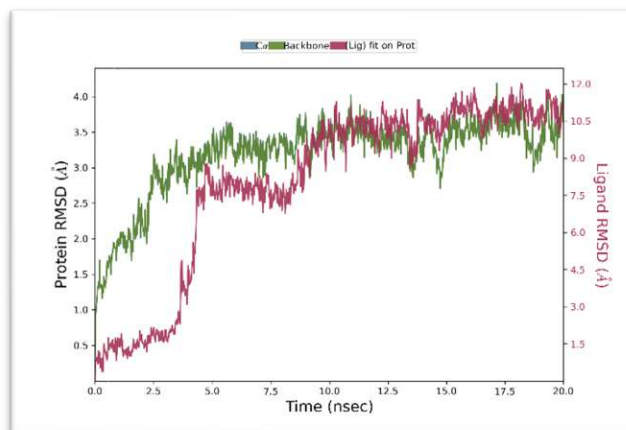
Ligand Properties: Throughout the simulation, the ligand RMSD remained low and stable between approximately 0.6 and 1.8 Å for the first 12 ns before showing a brief elevation to approximately 0.8–1.2 Å toward the latter part of the trajectory, indicating a minor conformational adjustment within the binding pocket without dissociation from the active site. The radius of gyration was consistently maintained between 4.25 and 3.8 Å throughout the full 20 ns, confirming that the ligand retained a compact and stable three-dimensional conformation during the simulation. Intramolecular hydrogen bonds fluctuated between 0 and 1 per frame, indicating that the ligand's hydroxyl groups were predominantly directed outward toward protein residues rather than forming internal bonds, which is favorable for intermolecular binding. The molecular surface area remained stable between 294 and 282 Å², the solvent accessible surface area ranged from 80 to 160 Å². The polar surface area was consistently maintained between 180 and 150 Å², reflecting a well-embedded polar ligand conformation within the JAK1 active site throughout the simulation.



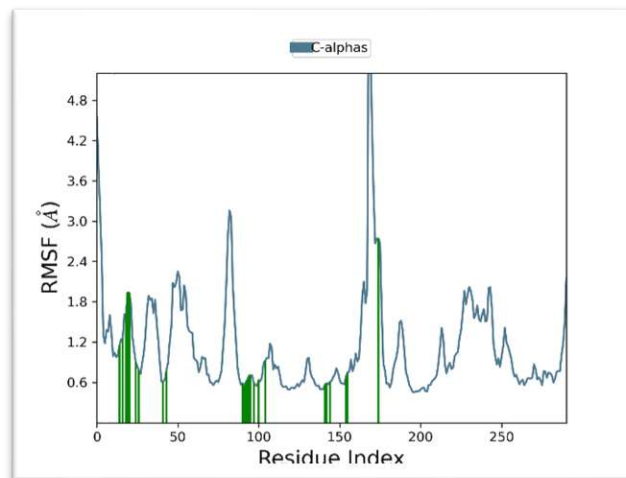
4) JAK2 with Hecogenin

The molecular dynamics simulation of the JAK2-HEC complex was conducted for 20 ns under NPT ensemble conditions at 300K temperature using Desmond. The system comprised of 37939 atoms along with 11026 water molecules and a net charge of zero

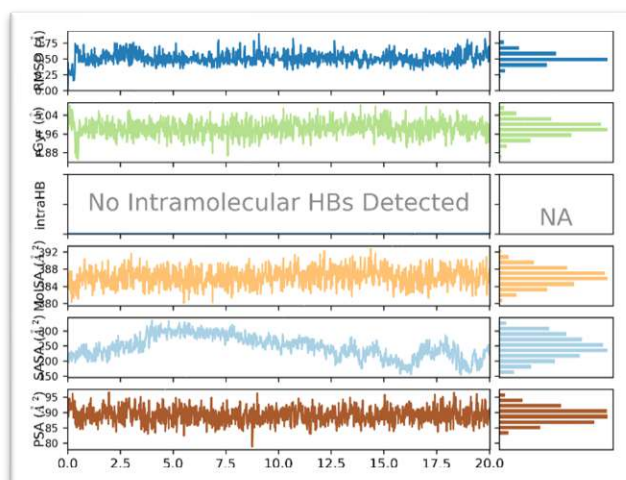
Structural Stability (RMSD): The protein backbone RMSD fluctuated between 1.00 and 4.00 Å throughout the 20 ns simulation. The system reached equilibrium. The ligand remained consistent throughout the simulation with a small peak at 3.5 Å suggesting the system reached equilibrium. The ligand RMSF ranged from 0.5 to 10.5 Å with peaks at 2.5 and 5 ns, with values comparable to or slightly exceeding the protein RMSD at various time points.



Residue Flexibility (RMSF): Protein RMSF analysis revealed that the majority of residues maintained fluctuations between 1.2 and 4.2 Å, with localized peaks reaching up to 3.0 Å and 4.8 Å at loop regions and terminal segments, consistent with the expected flexibility of unstructured protein regions. Ligand-contacting residues, indicated by green bars at approximately residue indices 30–55, 100, and 145–155, exhibited comparatively low fluctuation values, confirming that the binding site residues maintained structural stability throughout the simulation despite the overall conformational dynamics of the protein.



Ligand Properties: Throughout the simulation, the ligand maintained a stable RMSD of approximately 0.25 – 0.75 Å and a consistent radius of gyration between 4.88 and 5.04 Å, confirming a compact and stable bound conformation. The molecular surface area remained stable between 380 and 392 Å², the solvent accessible surface area ranged from 80 to 160 Å². The polar surface area was consistently maintained between 300 and 200 Å², reflecting a well-embedded polar ligand conformation within the JAK1 active site throughout the simulation.

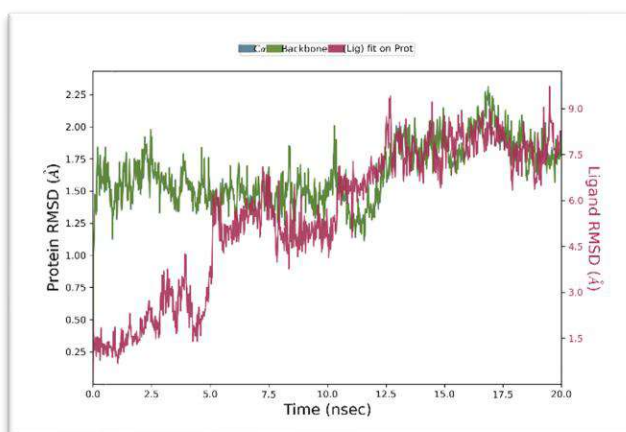


4.8.3. JAK3 Section:

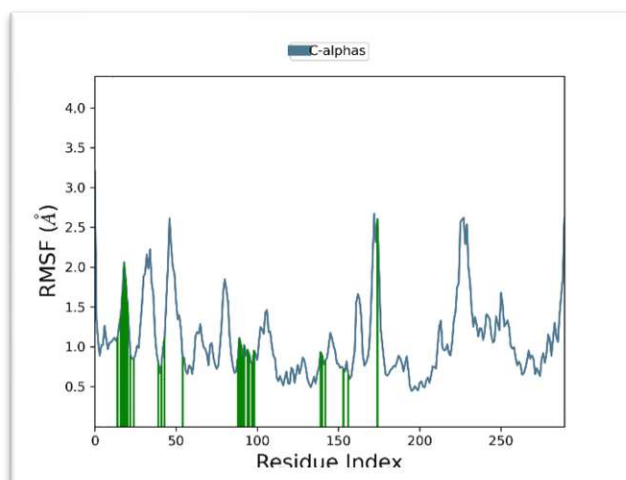
5) JAK3 with Hecogenin

The molecular dynamics simulation of the JAK3-HEC complex was conducted for 20 ns under NPT ensemble conditions at 300K temperature using Desmond. The system comprised of 35825 atoms along with 10378 water molecules and a net charge of zero.

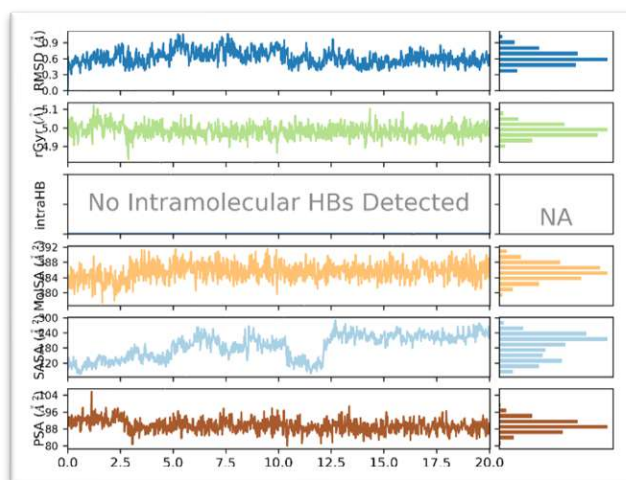
Structural Stability (RMSD): The protein backbone RMSD fluctuated between 0.75 and 2.00 Å throughout the simulation, remaining within the threshold limit of 1-3 Å for globular proteins. The ligand RMSD (Lig fit Prot) ranged from 1.0 to 4.5 Å without exhibiting a continuous upward drift, suggesting that the ligand remained associated with the binding pocket throughout the trajectory.



Residue Flexibility (RMSF): Protein RMSF analysis showed that the majority of residues fluctuated below 1.6 Å, with elevated peaks of up to 3.6 Å observed only at loop and terminal regions, consistent with expected protein dynamics. Residues in direct contact with the ligand, highlighted by green markers, displayed low fluctuation values below 1.2 Å, confirming binding site stability. Ligand RMSF analysis revealed atomic fluctuations between 0.5 and 1.6 Å.



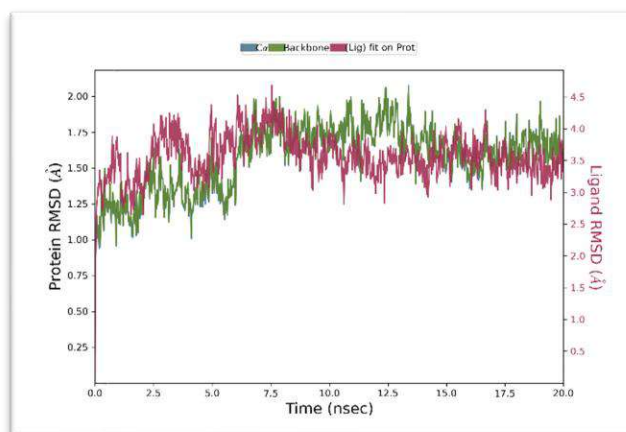
Ligand Properties: Throughout the simulation, the ligand maintained a stable RMSD of approximately 1.0–1.5 Å and a consistent radius of gyration between 5.5 and 6.1 Å, confirming a compact and stable bound conformation. The high polar surface area ranging from 360 to 420 Å² is consistent with the molecular formula C₂₉H₅₂O₁₅ and supports the observed hydrogen bonding profile. Intramolecular hydrogen bonds were observed intermittently at 0–2 per frame, indicating that the ligand's hydroxyl groups preferentially engage with protein residues rather than forming internal bonds.



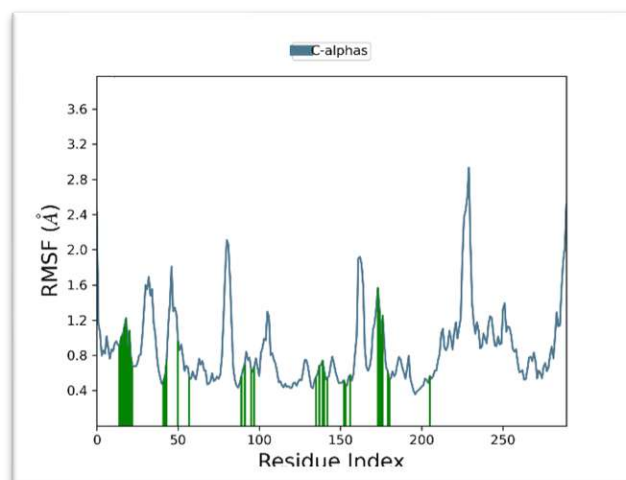
6) JAK3 with Verbacoside

The molecular dynamics simulation of the JAK3-HEC complex was conducted for 20 ns under NPT ensemble conditions at 300K temperature using Desmond. The system comprised of 35843 atoms along with 10377 water molecules and a net charge of zero.

Structural Stability (RMSD): The protein backbone RMSD fluctuated between 1.00 and 1.75 Å throughout the simulation, remaining within the threshold limit of 1-3 Å for globular proteins. The ligand RMSD (Lig fit Prot) ranged from 1.0 to 3.5 Å without exhibiting a continuous upward drift, suggesting that the ligand remained associated with the binding pocket throughout the trajectory.



Residue Flexibility (RMSF): Protein RMSF analysis showed that the majority of residues fluctuated below 2.8 Å, with elevated peaks of up to 2.0 Å and 2.8 Å observed only at loop and terminal regions, consistent with expected protein dynamics. Residues in direct contact with the ligand, highlighted by green markers, displayed low fluctuation values below 1.6 Å, confirming binding site stability. Ligand-contacting residues, indicated by green bars at approximately residue indices 30–55, 100, and 145–155, 175 and 200 exhibited comparatively low fluctuation values, confirming that the binding site residues maintained structural stability throughout the simulation despite the overall conformational dynamics of the protein.



Ligand Properties: Throughout the simulation, the ligand maintained a stable RMSD of approximately 1.0–1.5 Å and a consistent radius of gyration between 5.5 and 6.1 Å, confirming a compact and stable bound conformation. The high polar surface area ranging from 360 to 420 Å² is consistent and supports the observed hydrogen bonding profile. Intramolecular hydrogen bonds were observed intermittently at 0–2 per frame, indicating that the ligand's hydroxyl groups preferentially engage with protein residues rather than forming internal bonds.

5. Discussions

As mentioned in the introduction the importance of Eczema and its side effects on daily routines in affected individuals. Prior studies have noted the importance of JAK/STAT pathway in occurrence of eczema. FDA-Apporved drugs have been released into the market as JAK inhibitors which act againts the three isoforms of JAK (JAK1, JAK2 and JAK3) and have proved to show effect againts eczema. An intial objective of this research was to conduct studies to find the natural compunds that can be used as JAK inhibitors by using Molecular Docking and Molecular Simulation approach. The present study was designed to determine the effect of 14 natural compounds againts JAK1, JAK2 and JAK3 collectively. With respect to the first question it was found that, these natural compunds acting againts the JAK/STAT pathway to cure eczema might be used in treating cancers as it is also regulated through the same pathway. The results of this study show that, the three isoforms of JAK contain structural similarity in its Tyrosine Kinase 1 domain, as, it is where the ATP-binding pocket is conserved. This stated that the binding pocket is conserved with minor changes in it's protein constitution. On the question of JAK inhibitors, 14 natural compounds were screened using PyRx software. Diosmetin and Naringenin were selected againts JAK1. Hecogenin and Diosmetin were selected againts JAK2 and Hecogenin and Verbacoside were selected againts JAK3. Although for JAK3 Asthaxanthin had shown highest binding affinity but due to its poor ADMET studies it was not considered to be a stable ligand against JAK3. These 6 compounds were then docked with each of their isoforms using UCSF Chimera 1.19 and AutoDock Vina. The docking scores were also similar to the screening scores which indicated that the process was carried out perfectly. Another important finding was the data from Molecular Simulation studies. Among the six protein-ligand complexes subjected to 20 ns MD simulation, three complexes JAK3 with Verbacoside, JAK3 with Hecogenin, and JAK2 with Hecogenin demonstrated the most favorable stability profiles, warranting detailed discussion of their binding behavior and therapeutic implications. The JAK3-Verbacoside complex exhibited the most stable trajectory among all simulated systems, with protein backbone RMSD values confined between 1.00 and 1.75 Å throughout the 20 ns simulation, well within the acceptable range of 1–3 Å for globular proteins. This degree of structural stability suggests that Verbacoside binding does not impose significant conformational stress on the JAK3 kinase domain. The ligand RMSD remained between 1.0 and 3.5 Å without exhibiting a continuous upward drift, indicating stable residence within the ATP-binding pocket. Notably, no intramolecular hydrogen bonds were detected throughout the simulation, implying that the compound's multiple hydroxyl groups are entirely engaged in intermolecular contacts with the surrounding protein residues, thereby maximizing binding complementarity. This interaction profile is particularly relevant in the context of atopic dermatitis, as JAK3 is a critical mediator of IL-4 and IL-13 signaling through the JAK1-JAK3 heterodimer, and its selective inhibition has been demonstrated to attenuate the Th2-skewed inflammatory response characteristic of this condition. The next complex which demonstrated a great simulation data was the JAK3-Hecogenin complex. The rapid stabilization of the RMSD in the early phase of the trajectory suggests that Hecogenin achieves stable binding conformation relatively quickly, a desirable property for potential drug candidates as it implies a well-defined binding mode with minimal induced-fit reorganization of the protein. The ligand RMSF, confined between 0.5 and 1.6 Å across all heavy atoms, indicates that the rigid steroidal scaffold of Hecogenin undergoes minimal conformational sampling within the binding pocket, consistent with the low number of rotatable bonds characteristic of steroidal compounds. The absence of intramolecular hydrogen bonds throughout the simulation mirrors the findings observed in the JAK3-Verbacoside complex and similarly suggests that the hydroxyl substituents of Hecogenin are predominantly directed toward protein residues, facilitating intermolecular hydrogen bonding within the active site. Given that JAK3 signaling is implicated in the transduction of IL-31, the primary pruritogenic cytokine in atopic dermatitis, the stable inhibitory complex formed between Hecogenin and JAK3 may offer a mechanistic basis for alleviating both the inflammatory and itch components of this condition. The last complex which revealed a stable binding was the JAK2-Hecogenin complex. The JAK2-Hecogenin complex revealed a stable simulation profile characterized by protein backbone RMSD values between 1.00 and 4.00 Å, with convergence observed following the initial equilibration phase. While the RMSD range is comparatively wider than that observed in the JAK3 complexes, the absence of a sustained upward trend and the stabilization of

trajectory fluctuations around a thermal average are indicative of a system that achieved dynamic equilibrium within the simulation timeframe. Of particular significance is the exceptionally stable ligand conformation observed in this complex, with ligand RMSD values confined between 0.25 and 0.75 Å the lowest among all three lead complexes and a radius of gyration maintained consistently between 4.88 and 5.04 Å, confirming that Hecogenin adopted a highly stable and compact bound conformation within the JAK2 active site. From a therapeutic perspective, JAK2 plays an important role in transducing signals from IL-31 and thymic stromal lymphopoietin (TSLP), cytokines closely associated with the pruritic and barrier-disrupting phenotype of atopic dermatitis. The stable inhibitory interaction of Hecogenin with JAK2, combined with its demonstrated stability in the JAK3 complex, raises the possibility of a multi-target inhibitory profile for this natural compound. Such pan-JAK activity, targeting both JAK2 and JAK3 simultaneously, could provide broader suppression of the cytokine-driven inflammatory cascade in atopic dermatitis.

6. Conclusion

In this investigation, the aim was to assess potential JAK inhibitors from natural compounds through a systematic computational screening approach involving molecular docking and molecular dynamics simulation. Given the well-established role of the JAK-STAT signaling pathway in mediating the cytokine-driven inflammatory cascade of atopic dermatitis, and the limitations associated with currently approved JAK inhibitors in terms of cost, systemic side effects, and safety concerns, this study aimed to explore natural compounds as safer and structurally diverse alternatives. A library of natural compounds was screened against JAK1, JAK2, and JAK3 using molecular docking, and the most promising candidates were subjected to 20 ns molecular dynamics simulation to evaluate the stability and persistence of their binding interactions under physiological conditions. The top-ranked compounds demonstrated high binding energies and RMSD values with key interactions involving conserved active site residues critical for kinase activity. Among the screened compounds, Diosmetin, Hecogenin, Verbacoside, and Naringenin emerged as the most promising candidates based on their docking scores, interaction profiles, and predicted ADMET properties, and were subsequently advanced to molecular dynamics simulation for further evaluation. Molecular dynamics simulation over 20 ns was performed for six protein-ligand complexes, encompassing JAK1 with Diosmetin and Naringenin, JAK2 with Diosmetin and Hecogenin, and JAK3 with Hecogenin and Verbacoside. All six complexes demonstrated overall system stability throughout the simulation trajectory, with protein backbone RMSD values reaching equilibration and ligand RMSD profiles showing no sustained divergence from the binding site, confirming that none of the candidate compounds dissociated from their respective active sites during the simulation period. Secondary structure analysis confirmed the preservation of α -helical and β -strand content across all protein targets, indicating that none of the ligands induced significant structural disruption of the JAK kinase domains. Among the six complexes, three demonstrated particularly outstanding stability profiles. The JAK3-Verbacoside complex exhibited the narrowest protein backbone RMSD range of 1.00–1.75 Å, representing the most structurally stable complex of the study. The JAK3-Hecogenin complex maintained RMSD values between 0.75 and 2.00 Å with rapid equilibration and tight ligand conformation. The JAK2-Hecogenin complex was distinguished by the most stable ligand geometry of all complexes, with ligand RMSD confined between 0.25 and 0.75 Å and a highly consistent radius of gyration, confirming exceptional conformational stability of Hecogenin within the JAK2 active site. Based on the integrated analysis of molecular docking scores, protein-ligand interaction profiles, and molecular dynamics simulation parameters including RMSD, RMSF, secondary structure integrity, protein-ligand contacts, and ligand properties, two natural compounds emerged as the most promising lead candidates from this study. Hecogenin, a steroidal sapogenin of natural origin, demonstrated stable and persistent binding to both JAK2 and JAK3, suggesting a multi-target inhibitory potential that could simultaneously suppress multiple cytokine signaling axes implicated in atopic dermatitis, including IL-31 mediated pruritus via JAK2 and IL-4/IL-13 driven Th2 inflammation via JAK3. Verbacoside, a phenylethanoid glycoside, demonstrated selective and highly stable binding to JAK3, with the most favorable protein conformational stability observed in the entire study and a complete absence of intramolecular hydrogen bonding, indicating full engagement of its hydroxyl groups with active site residues. This study contributes to the growing body of literature exploring natural products as sources of novel kinase inhibitors and demonstrates the utility of integrated computational approaches in the early-stage drug discovery pipeline. The identification of Hecogenin and Verbacoside as potential JAK inhibitors through systematic *in silico* screening provides a rational basis for their further investigation as candidates for atopic dermatitis therapy.

7. Future Prospects

In a clinical landscape dominated by expensive biologic agents and JAK inhibitors carrying black box safety warnings, the exploration of natural compounds with established safety profiles and structural diversity represents a valuable complementary approach to developing more accessible and tolerable treatment options for patients with atopic dermatitis. It is important to acknowledge that the findings of this study are based entirely on computational methods and represent predictive rather than experimentally validated results. Molecular docking and molecular dynamics simulation, while powerful tools for evaluating binding interactions, are subject to limitations including force field approximations, fixed protonation states, and the absence of membrane environment and cellular context. The 20 ns simulation timescale, while sufficient for initial stability assessment, may not fully capture the conformational dynamics that occur over longer biological timescales. The stability of the identified protein-ligand complexes, in future investigations must be analysed by extending the simulation timescale to 100–500 ns or beyond to capture slower conformational dynamics and more comprehensively evaluate the long-term binding stability of Hecogenin and Verbacoside within their respective JAK binding pockets. Furthermore, the ADMET profiles predicted in this study are based on in silico models and require experimental validation through in vitro pharmacokinetic and toxicity assays. Quantitative structure-activity relationship modeling could further accelerate the identification of optimized derivatives with superior JAK inhibitory potency. The absence of binding free energy calculations such as MM-GBSA or MM-PBSA represents a further limitation that future work should address to provide more rigorous quantitative ranking of the identified lead compounds. Given that Hecogenin demonstrated stable binding to both JAK2 and JAK3 in the present study, future investigations should employ network pharmacology approaches to comprehensively map the multi-target interactions of this compound within the broader inflammatory signaling network relevant to atopic dermatitis. Cell-based assays should be conducted to evaluate the ability of Hecogenin and Verbacoside to suppress JAK-STAT pathway activation in biologically relevant cellular models of atopic dermatitis. Cytotoxicity assays including MTT and LDH assays should be performed in parallel to establish the therapeutic window of each compound and ensure an acceptable safety profile at pharmacologically relevant concentrations. In conclusion, this study provides a comprehensive computational framework demonstrating that Hecogenin and Verbacoside are promising natural compound leads targeting the JAK-STAT pathway in atopic dermatitis, and their further experimental validation may contribute meaningfully to the development of safer, more accessible, and naturally derived JAK inhibitors for this prevalent skin condition.

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